Higher order comoments of multifactor models and asset allocation☆

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Abstract

Accurate estimates of the higher order comoments are needed in asset allocation. We derive explicit formulas for the higher order comoments under the assumption that stock returns are generated by a multifactor model and show that this assumption leads to a substantial reduction in the number of parameters to estimate compared to the traditional approach. An out-of-sample analysis of the performance of portfolio allocation criteria that depend on the higher order comoments illustrates the usefulness of the proposed methodology.

Keywords: factor models, higher order comoments, portfolio selection
JEL: G110, C580

1. Introduction

The distribution of asset returns is often asymmetric and heavy tailed. If there is no estimation error, most investors would be willing to sacrifice expected return and/or accept a higher volatility in exchange for a higher skewness and lower kurtosis leading to a lower downside risk (e.g. Ang *et al.* (2006), Harvey and Siddique (2000)). This trade-off between positive preferences for odd moments (mean, skewness) and negative preferences for even moments (variance, kurtosis) can be conveniently summarized into a single objective function using a Taylor expansion of the expected utility function as objective (as e.g. in

☆The authors would like to thank Jorn De Boeck, Giang Nguyen, and Joshua Ulrich, and the doctiris program from the Brussels region for support, as well as participants of the 5th R/Finance conference in Chicago (2014). Wanbo Lu's research is sponsored by the National Science Foundation of China (71101118) and the Program for New Century Excellent Talents in University (NCET-13-0961).

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Preprint submitted to SSRN *June 16, 2014*

Jondeau and Rockinger (2006) ar Martellini and Ziemann (2010)) or a portfolio downside risk objective based on the Cornish-Fisher expansion (e.g. Boudt *et al.* (2013)).

The important caveat is that portfolio moments need to be estimated, and that, without restrictions on the data generating process, the number of parameters to estimate becomes quickly very large when the dimension of the investment universe increases. This curse of dimensionality makes the unrestricted estimators of the first four (co)moments almost infeasible for moderately large dimensions. Suppose e.g. that we have a universe of 20 assets, then the number of unique elements in the covariance, coskewness and cokurtosis is 210, 1540 and 8555, respectively. This is clearly an excessive number of parameters compared to the number of observations that are available in realistic applications. As noted by Michaud (1998), the portfolio optimizer acts like an "error maximizer" and amplifies the estimation errors even further when optimizing the portfolio weights and leads to optimized portfolios that are often not well-diversified (Green and Hollifield (1992)).

In this article, we avoid the curse of dimensionality by assuming that the asset returns are generated by multifactor model. The total number of unique elements in the covariance, coskewness and cokurtosis matrix of 20 assets is 83, 112 and 151 for 1, 2 and 3 factors, respectively. The only related papers are Martellini and Ziemann (2010) recommending to use a single factor model assumption and Ghalanos *et al.* (2014) who model the higher order moments of asset returns assuming the returns can be rewritten as a linear combination of independent factors. Our approach is more general and decomposes the return vector into a linear combination of a lower dimensional set of possibly dependent factors and a vector of residual terms that can be interpreted as independent idiosyncratic factors.

We illustrate the usefulness of the multifactor approach to higher order comoments in an international portfolio context where the investor allocates with the purpose to maximize his expected utility. The universe consists of four equity benchmarks, nine bond indices and five commodity indices. We find that accounting for the higher order moments using a multifactor approach increases out-of-sample average returns, decreases portfolio standard deviations and leads to an important reduction in the portfolio downside risk.

In what follows, we first describe in Section 2 the general higher order moment estimation framework that we propose. This includes our key contribution, which is the derivation of the explicit formula for the higher order comoments when the asset returns are generated by a multifactor model. In Section 3 we then study the out-of-sample performance of portfolios that use these higher order comoments. The major findings are summarized in the conclusions.

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2. The higher order comoments of the multifactor model and their application in asset allocation

2.1. Asset allocation, higher order comoments and the factor model

Besides the forecasted mean return, the key input parameters for the portfolio decision that we study are the covariance, coskewness and cokurtosis matrix of the $N-$ dimensional return vector r with mean μ_r, i.e. the comoments corresponding to (i) the products of two returns, i.e. the covariance of assets i and j:

$$\sigma_{i,j} = E[(r_{(i)} - \mu_{r(i)})(r_{(j)} - \mu_{r(j)})], \tag{1}$$

(ii) the products of three returns, i.e. the coskewness of assets i, j and k:

$$\phi_{i,j,k} = E[(r_{(i)} - \mu_{r(i)})(r_{(j)} - \mu_{r(j)})(r_{(k)} - \mu_{r(k)})], \tag{2}$$

and (iii) the products of four returns, i.e. the cokurtosis of assets i, j, k and l:

$$\psi_{i,j,k,l} = E[(r_{(i)} - \mu_{r(i)})(r_{(j)} - \mu_{r(j)})(r_{(k)} - \mu_{r(k)})(r_{(l)} - \mu_{r(l)})]. \tag{3}$$

It will reveal useful to stack all these comoments into a $N \times N$ covariance matrix Σ, $N \times N^2$ coskewness matrix Φ and $N \times N^3$ cokurtosis matrix Ψ of the return vector r, i.e.:

$$\begin{aligned} \Sigma &= E[(r - \mu_r)(r - \mu_r)'], \\ \Phi &= E[(r - \mu_r)(r - \mu_r)' \otimes (r - \mu_r)'], \\ \Psi &= E[(r - \mu_r)(r - \mu_r)' \otimes (r - \mu_r)' \otimes (r - \mu_r)'], \end{aligned} \tag{4}$$

where $\otimes$ denotes the Kronecker product. These comoments allow us to easily calculate the k-th portfolio return moment for a portfolio with weights w:

$$\begin{aligned} m_2(w) &= E[(w'(R - \mu_R))^2] = w'\Sigma w, \\ m_3(w) &= E[(w'(R - \mu_R))^3] = w'\Phi(w \otimes w), \\ m_4(w) &= E[(w'(R - \mu_R))^4] = w'\Psi(w \otimes w \otimes w). \end{aligned} \tag{5}$$

The challenge is to estimate these comoment matrices Σ, Φ and Ψ. As shown in Table 1, for moderately sized portfolios of size $N = 20$, it requires to estimate 10,605 parameters. To avoid this curse of dimensionality, the solution we provide is to impose more structure

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Table 1: Number of elements to estimate under the unrestricted and multifactor model approach.

	Unrestricted	Factor model with K factors		
Σ	$N(N+1)/2$	$N(K+1)+K(K+1)/2$		
Φ	$N(N+1)(N+2)/6$	$N(K+1)+K(K+1)(K+2)/6$		
Ψ	$N(N+1)(N+2)(N+3)/24$	$N(K+2)$ $+K(K+1)/2$ $+K(K+1)(K+2)(K+3)/24$		
Total	$N(N+1)/2$ $+N(N+1)(N+2)/6$ $+N(N+1)(N+2)(N+3)/24$	$N(K+3)$ $+[(1+(K+2)/3$ $+(K+2)(K+3)/12]K(K+1)/2$		
	Total	$K=1$	$K=2$	$K=3$
$N=5$	120	23	37	61
$N=20$	10,605	83	112	151
$N=100$	4,598,025	403	512	631

on the data through a factor model, under which it is assumed that the variation in asset returns is driven by multiple common factors and idiosyncratic factors that are specific to each asset.

Three types of factor models exist. In the macroeconomic factor model, factors are observable macro-financial variables. Under the fundamental factor model, factors are created from observable asset characteristics. Finally, in statistical factor models, factors are unobservable and extracted from the asset returns. Connor (1995) compares the explanatory power of the three types of factor models for security market returns and finds that the statistical and fundamental factor models outperform the macroeconomic model. In the application on asset allocation, we will use statistical factors, as it is not obvious to find a parsimonious set of fundamental factors.

All three types of factor models can be represented in the same general form. To introduce this notation, suppose that K observable factors are identified as being influential for the portfolio variability, and that, at a given frequency, the asset returns $r_t = (r_{1t}, \cdots, r_{Nt})'$ and the factors $f_t = (f_{1t}, \cdots, f_{Kt})'$ are recorded. The asset returns are assumed to depend linearly on the factors, whereby the variation in the asset returns that is not explained by the factors (denoted as e_t), is assumed to be independent of each of the factors and also to be independent across assets. In matrix notation, the system is given by:

$$r_t = a + Bf_t + e_t, \tag{6}$$

where $e_t = (e_{1t}, \cdots, e_{Nt})'$ is the $N \times 1$ vector of independent asset specific factors and B is the $N \times K$ matrix of factor loadings (also called factor beta's or factor exposures) of the N assets on the K factors.

2.2. Comoments under the linear factor model

Let S be the $K \times K$ covariance matrix of the K factors, G the $K \times K^2$ coskewness matrix of the K factors and P their $K \times K^3$ cokurtosis matrix:

$$\begin{aligned}
\mu_f &= E[f_t], \\
S &= E[(f_t - \mu_f)(f_t - \mu_f)'], \\
G &= E[(f_t - \mu_f)(f_t - \mu_f)' \otimes (f_t - \mu_f)'], \\
P &= E[(f_t - \mu_f)(f_t - \mu_f)' \otimes (f_t - \mu_f)' \otimes (f_t - \mu_f)'].
\end{aligned} \tag{7}$$

We rewrite the comoment matrices Σ, Φ and Ψ as the sum of the comoment of the return explained by the factor (i.e. the comoment of Bf_t) and a residual matrix denoted by Δ, Ω and Υ:

$$\begin{aligned}
\Sigma &= BSB' + \Delta, \\
\Phi &= BG(B \otimes B') + \Omega, \\
\Psi &= BP(B' \otimes B' \otimes B') + \Upsilon.
\end{aligned} \tag{8}$$

Because of the assumption that the unexplained asset return variation e_t is independent of the factors, Δ is a diagonal matrix with i-th diagonal element equal to the variance of the $i-$th error term and Ω is a $N \times N^2$ matrix of zeros except for the i, j elements where $j = (i-1)N + l$, which is corresponding to the expected third moment of the idiosyncratic factors.

The definition of Υ is slightly more complex. Like the other residual matrices, it consists mostly of zeros, except for the cokurtosis elements corresponding to the decomposition of: the kurtosis of one asset, the cokurtosis between two assets and the cokurtosis between 3 assets. For the kurtosis of one asset (i.e. $i = j = k = l$), the corresponding element in Υ should be:

$$6b_i'Sb_iE[e_{it}^2] + E[e_{it}^4]. \tag{9}$$

For the cokurtosis between two assets when $(i = j = k)$ and $l \neq i$, the corresponding element in Υ should be:

$$3b_i'Sb_lE[e_{it}^2], \tag{10}$$

and analogously for $(i = j = l)$ and $k \neq l$, or $(j = k = l)$ and $i \neq k$. When $(i = j) \neq (k = l)$, the corresponding element in Υ should be:

$$b_i' S b_i E[e_{kt}^2] + b_k' S b_k E[e_{it}^2] + E[e_{it}^2] E[e_{kt}^2], \tag{11}$$

and similarly for $(i = l) \neq (j = k)$ and $(i = k) \neq (j = l)$. Finally, for the cokurtosis between 3 assets, i.e. $i = j$ and $k \neq i$, $k \neq l$, $i \neq l$, the corresponding element in Υ should be:

$$b_k' S b_l E[e_{it}^2], \tag{12}$$

and analogously for similar combinations. See Appendix for the proof.

2.3. Estimation

There is considerable evidence of autocorrelation in the variance, skewness and kurtosis (see e.g. Jondeau and Rockinger (2003) and Leon *et al.* (2005)). Several approaches exist to take this time-variation in the higher order moments into account. In the application, we will take the rolling estimation sample approach and avoid making assumptions on the functional form of the dynamic linkage between future and past returns. Alternative parametric approaches are based on garch-type of modelling of the dynamic comoments (e.g. Boudt *et al.* (2013); Ghalanos *et al.* (2014)) or accounting for regime switches in the return distribution (e.g. Bae *et al.* (2014)). The factor exposures are estimated by the ordinary least squares estimator and population moments are estimated by the corresponding sample averages.

3. Empirical illustration

3.1. Data and portfolio methods

The proposed methodology has important applications in the design of optimal asset allocation portfolios. We illustrate this for a realistic universe of four equity benchmarks (Europe, North America, Pacific and Emerging Markets), eight bond indices (corporate developed high yield index in EUR and USD, corporate developed investment grade index in EUR, corporate emerging investment grade in USD, sovereign developed investment grade in USD, EUR, JPY and sovereign developed and emerging in USD) and five commodity indices (agriculture, energy, industrial metals, livestock and precious metals).

The cumulative return evolution of each of the asset classes over the period 1999-2012 is shown in Figure 1. Besides the differences in volatility and return over the period, the

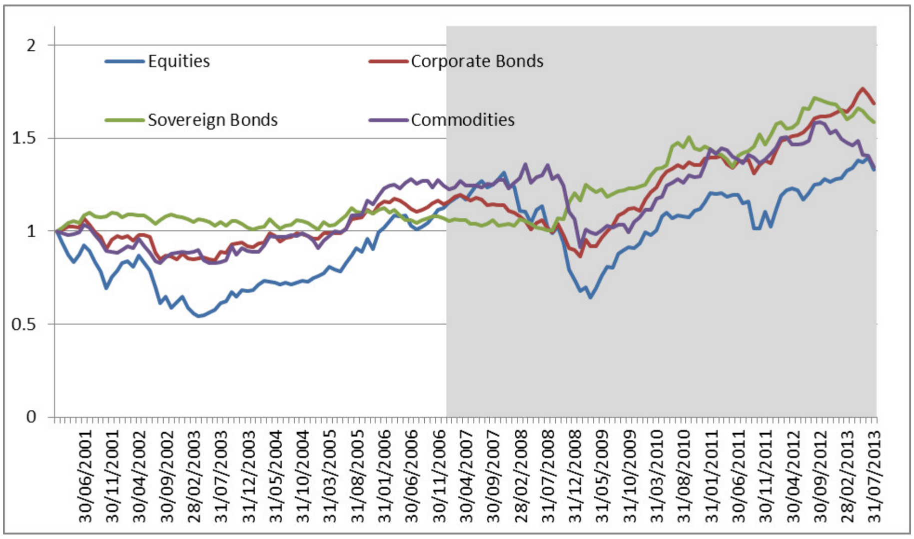

Figure 1: Monthly cumulative level of equal-weighted equity, equal-weighted sovereign bonds, equal-weighted corporate bonds and equal-weighted commodities over the period February 2001-July 2013 (in EUR). The grey area indicates the out-of-sample evaluation period used.

graph clearly shows the diversification potential across the different investment universes. The shaded area in Figure 1 corresponds to the out-of-sample evaluation period used to compare the different portfolio allocation methodologies.

Like in Martellini and Ziemann (2010) we will assume that the investor maximizes the expected value of the fourth order expansion of the Constant Relative Risk Aversion (CRRA) utility function with risk aversion parameter γ and we neutralize the effect of the mean by assuming it to be zero. This then leads to the following expected utility function:

$$EU_{\gamma}(w) = -\frac{\gamma}{2}m_{(2)}(w) + \frac{\gamma(\gamma+1)}{6}m_{(3)}(w) - \frac{\gamma(\gamma+1)(\gamma+2)}{24}m_{(4)}(w), \tag{13}$$

where $m_{(2)}(w)$, $m_{(3}(w)$ and $m_{(4)}(w)$ is the second, third and fourth portfolio moment as defined in (5). Like in Martellini and Ziemann (2010) we will consider $\gamma = 10$ as our base case but will also try the more common choice of $\gamma = 5$. In order to avoid that the portfolio's risk exposure is concentrated in the least risky assets, we additionally impose the equal risk contribution (ERC) diversification constraint that all asset classes contribute equally to portfolio risk, based on the standard Euler decomposition. The properties of imposing the ERC constraint in variance based portfolios have been studied in Maillard *et al.* (2010), among others. For asymmetric distributions, a downside risk measure is preferred by many investors. We will use the modified Expected Shortfall estimator proposed by (Boudt *et* *al.* (2008) in which the third order Cornish-Fisher expansion is used to account for the skewness and kurtosis when estimating the portfolio risk. More precisely, the estimator for the expected shortfall at the level α (e.g. 5 per cent) is given by:

$$ES_\alpha(w) = -w'\mu + \sqrt{m_{(2)}(w)} \times \frac{1}{\alpha}[a_\alpha + b_\alpha k(w) + c_\alpha s(w) + d_\alpha s^2(w)]. \tag{14}$$

with $a_\alpha, b_\alpha, c_\alpha$, and d_α numbers that depend on the choice of the loss level α (see Boudt *et al.* (2008)) and $s(w)$ and $k(w)$ are the portfolio skewness and excess kurtosis respectively: $s(w) = m_{(3)}(w)/m_{(2)}^{3/2}(w)$ and $k(w) = m_{(4)}(w)/m_{(2)}^2(w) - 3$.

The optimizer of the asset allocation problem needs to search for the portfolio weight vector w that minimizes an objective function which (besides w) depends non-linearly on the higher order comoment matrices Σ, Φ and Ψ under a non-linear risk diversification constraint that also depend on Σ, Φ and Ψ, as well as a full investment constraint and box constraints on the portfolio weights. We will solve this problem using the heuristic optimization method Differential Evolution in which the box and full investment constraints are directly imposed through a mapping of the generated populations into the feasible space, while the equal risk contribution constraint is imposed as a penalty in the objective function. We use the implementation of Storn and Price (1997), as implemented in the R package of Ardia *et al.* (2013).

3.2. Results

Table 2 compares the January 2007-July 2013 out of sample returns for the weekly rebalanced portfolio methodologies, which either aim to minimize variance, or to maximize the CRRA expected utility with risk aversion parameter γ (5 and 10), under the risk diversification constraint that all three major asset classes contribute equally to portfolio variance or to the 95% portfolio expected shortfall. Portfolios are fully invested and short sales positions are not allowed. This leads to six constrained portfolio objectives. Each objective depends on the estimated comoments, for which we evaluate three options: the sample estimator, the single factor model based estimate and a 3-factor approach. In total we have thus 18 portfolio allocation schemes. The portfolios are optimized using the Differential Evolution global optimization method, as described above. The comoments are estimated using weekly returns in local currency and a rolling sample of six years (i.e. 312 observations).

The standard approach is in the first row of Table 2. It consists in minimizing the portfolio variance under the equal variance contribution constraint. Comparing this method with the other 17 portfolio allocation methodologies, the overall picture appears to be that

Table 2: Monthly returns analysis of weekly rebalanced minimum variance/maximum expected utility strategies under equal variance/expected shortfall contribution constraint over the period January 2007-July 2013 (in EUR).

Objective/ Risk Contribution	Estimator	Ann. return	Annualized standard deviation	Skewness	Excess kurtosis	95% Historical VaR	Max Drawdown
Min Variance/	Sample	1.02%	10.69%	-1.60	9.02	-5.60%	-25.52%
Variance	Single factor	3.02%	7.54%	0.49	1.67	-2.89%	-8.99%
	Multifactor	3.48%	7.84%	0.71	1.73	-2.83%	-18.56%
Min Variance/	Sample	5.08%	8.46%	0.17	0.90	-3.38%	-16.45%
E.Shortfall	Single factor	5.54%	7.90%	0.40	0.55	-2.96%	-7.86%
	Multifactor	4.70%	8.35%	0.33	0.52	-3.27%	-12.40%
Max E.Utility/	Sample	3.36%	8.32%	0.06	2.98	-3.27%	-9.89%
Variance	Single factor	5.85%	8.57%	0.64	0.70	-3.04%	-10.26%
($\gamma = 5$)	Multifactor	4.23%	8.73%	1.06	3.41	-3.44%	-13.27%
Max E.Utility/	Sample	5.47%	7.07%	0.27	0.77	-2.68%	-9.17%
E.Shortfall	Single factor	4.43%	9.59%	-0.88	6.75	-4.40%	-18.24%
($\gamma = 5$)	Multifactor	5.21%	9.67%	-0.98	6.88	-4.43%	-13.31%
Max E.Utility/	Sample	3.64%	8.03%	0.38	0.29	-3.19%	-11.65%
Variance	Single factor	1.80%	7.97%	0.01	0.57	-3.55%	-11.67%
($\gamma = 10$)	Multifactor	4.53%	8.90%	0.27	-0.18	-3.59%	-11.87%
Max E.Utility/	Sample	4.82%	9.05%	-0.05	0.38	-3.86%	-11.99%
E.Shortfall	Single factor	4.49%	8.25%	0.26	0.91	-3.28%	-10.27%
($\gamma = 10$)	Multifactor	9.47%	9.35%	1.43	3.32	-2.25%	-8.06%

the following two actions tend to increase the out-of-sample return and reduce the portfolio risk: (i) first, the estimation of comoments under a multifactor specification; (ii) second, the inclusion of higher order comoments in the objective function and risk diversification constraint. These effects are expected and in line with prior research that emphasized the risk of noise optimization when using the simple sample average-based estimates of the comoments in portfolio optimization (e.g. Jagannathan and Ma (2003), Ledoit and Wolf (2003), Martellini and Ziemann (2010)). Jondeau and Rockinger (2006) argued that, while the mean-variance criterion provides a good approximation of the expected utility maximization under moderate non-normality, it may be ineffective under large departure from normality, as the one in our sample. Because of this non-normality, risk diversification constraints based on expected shortfall are more appropriate than risk constraints based on the portfolio variance, as they take the non-linear dependence across asset returns into account.

For our sample, the best risk-adjusted portfolio performance is observed for the maximum expected utility objective with risk aversion parameter γ=10 and diversification imposed through an equal contribution to the expected shortfall constraint, whereby the comoments are estimated under the multifactor approach. This scenario shows an annualized return of 9.47%, an annualized volatility of 9.35%, positive skewness and a 95% portfolio VaR of 2.25%. Its maximum drawdown (8.06%) is less than the maximum drawdown on the least risky asset of the universe (Sov Developed IG, EUR: 8.46% maximum drawdown) at a substantially higher return (9.47% instead of 5.28%), and, importantly, because of the diversification constraint, it has a much smaller idiosyncratic risk.

In our out-of-sample analysis we thus confirm the result of Martellini and Ziemann (2010) and Jondeau and Rockinger (2012) that accounting for higher order moments can substantially improve portfolio performance. They showed this for equity portfolios, while our application is on asset allocation. Like in Martellini and Ziemann (2010), the size of the universe is relatively large (17 assets), explaining the need of structured estimates in order to increase the efficiency of the estimates and the resulting portfolio weights.

4. Conclusion

We propose a novel methodology to account for non-normality and the risk of "error maximization" when optimized portfolios take the classical higher order comoment estimates as input parameter. The building block of the proposed methodology is the assumption of a multifactor model generating the asset returns. This assumption allows for parsimoniously estimating the higher order comoments. Parsimony is achieved by striking a balance between generality (multiple factors) and the number of parameters to estimate. It extends the single factor approach in Martellini and Ziemann (2010) and the independent factor model in Ghalanos *et al.* (2014). This is the key contribution of this article. We further illustrate the usefulness of the method in an out-of-sample portfolio analysis for a heterogeneous universe of 17 equity, bonds and commodity indices over the period January 2007-July 2013.

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Appendix A. Proofs of decomposition higher order comoments

For notational convenience and without loss of generality, we assume here the returns and factors to be de-meaned such that $r_{it} = b_i' f_t + e_{it}$. Because we assume independence between the factors and residual terms, we have that, for $j \neq k$, and powers p and q: $E[(b_j' f_t)^p e_{jt}^q] = E[(b_j' f_t)^p] E[e_{jt}^q]$. The independence across residual terms implies further that, for $j \neq k$: $E[e_{jt}^p e_{kt}^q] = E[e_{jt}^p] E[e_{kt}^q]$. These independence assumptions allow us to simplify greatly the higher order comoments, as we show next.

We first consider the elements in the coskewness matrix $\Phi = BG(B' \otimes B') + \Omega$. We have that for the skewness of asset i:

$$\Phi_{i,i,i} = E[(b_i' f_t + e_{it})^3] = E[(b_i' f_t)^3 + 3(b_i' f_t)^2 e_{it} + 3(b_i' f_t) e_{it}^2 + e_{it}^3] = b_i' G(b_i \otimes b_i) + E[e_{it}^3].$$

Hence the residual element in Ω is $E[e_{it}^3]$. All other elements in Ω are 0, since the coskewness only comes from the common factor exposures:

$$\begin{aligned}
\Phi_{i,i,j} &= E[(b_i' f_t + e_{it})^2 (b_j' f_t + e_{jt})], \\
&= E[(b_i' f_t)^2 (b_j' f_t) + (b_i' f_t)^2 e_{jt} + 2(b_i' f_t) e_{it} (b_j' f_t) + 2(b_i' f_t) e_{it} e_{jt} + e_{it}^2 (b_j' f_t) + e_{it}^2 e_{jt}], \\
&= b_i' G(b_i \otimes b_j),
\end{aligned}$$

for $i \neq j$, and, for $i \neq j$, $j \neq k$, and $i \neq k$:

$$\Phi_{i,j,k} = E[(b_i' f_t + e_{it})(b_j' f_t + e_{jt})(b_k' f_t + e_{kt})] = b_i' G(b_j \otimes b_k).$$

We now consider the cokurtosis matrix $\Psi = BP(B' \otimes B' \otimes B') + \Upsilon$. The kurtosis of asset i is:

$$\begin{aligned}
\Psi_{i,i,i,i} &= E[(b_i' f_t + e_{it})^4] = E[(b_i' f_t)^4 + 4(b_i' f_t)^3 e_{it} + 6(b_i' f_t)^2 e_{it}^2 + 4(b_i' f_t) e_{it}^3 + e_{it}^4], \\
&= b_i' P(b_i \otimes b_i \otimes b_i) + 6 b_i' S b_i E[e_{it}^2] + E[e_{it}^4].
\end{aligned}$$

The result in (9) then follows. Then for the cokurtosis element, where $(i = j = k)$ and $l \neq i$:

$$\begin{aligned}
\Psi_{i,i,i,l} &= E[(b_i' f_t + e_{it})^3 (b_l' f_t + e_{lt})] = E[((b_i' f_t)^3 + 3(b_i' f_t)^2 e_{it} + 3(b_i' f_t)^2 e_{it}^2 + e_{it}^3)(b_l' f_t + e_{lt})], \\
&= b_i' P(b_i \otimes b_i \otimes b_j) + 3E[(b_i' f_t)(b_l' f_t) e_{it}^2] = b_i' P(b_i \otimes b_i \otimes b_j) + 3 b_i' S b_l E[e_{it}^2].
\end{aligned}$$

This proves the result in (10). We obtain (11) from:

$$\begin{aligned}
\Psi_{i,i,k,k} &= E[(b_i' f_t + e_{it})^2 (b_k' f_t + e_{kt})^2], \\
&= E[((b_i' f_t)^2 + 2(b_i' f_t)^2 e_{it} + e_{it}^2)((b_k' f_t)^2 + 2(b_k' f_t)^2 e_{kt} + e_{kt}^2)], \\
&= b_i' P(b_i \otimes b_k \otimes b_k) + b_i' S b_i E[e_{kt}^2] + b_k' S b_k E[e_{it}^2] + E[e_{it}^2] E[e_{kt}^2].
\end{aligned}$$

Finally, (12) is obtained using:

$$\Psi_{i,i,k,l} = E[(b_i' f_t + e_{it})^2 (b_k' f_t + e_{kt})(b_l' f_t + e_{lt})]$$

$$= E[((b_i' f_t)^2 + 2(b_i' f_t)^2 e_{it} + e_{it}^2)((b_k' f_t) + e_{kt})((b_l' f_t) + e_{lt})] = b_i' P(b_i \otimes b_k \otimes b_l) + b_k' S b_l E[e_{it}^2],$$

where $i \neq k$, $k \neq l$, and $i \neq l$. For all other elements that have no common index:

$$\Psi_{i,j,k,l} = E[(b_i' f_t + e_{it})(b_j' f_t + e_{jt})(b_k' f_t + e_{kt})(b_l' f_t + e_{lt})] = b_i' P(b_i \otimes b_k \otimes b_l).$$